COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	10/9/2026	**DATE SUBMITTED:**	10/9/2026
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ACTIVITY NO. 5.2 Structures

1. PROCEDURE OUTPUT

What procedures did you perform?

EXPLANATION: The program first starts by initiating a structure for books wherein it contains their own respective variables. After that, the program then initializes 2 variables for the structure Books, which was then filled with values by the programmer. Eventually the program printed the values with the corresponding book variable in the Books structure which was accessed by using the dot operator function.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after

.

Create a structure named Rectangle to store a rectangle's length and width. Write a program that uses the structure to compute and display the area and perimeter of the rectangle.

INPUT

CPE11S4-DOMINGO > STRUCT2HOA.cpp > ...

```cpp
#include <iostream>
#include <string>
using namespace std;

struct Rectangle {
    double l;
    double w;
};


int main () {
    char decision;
    Rectangle rec1;
    cout << "Rectangle." << endl;
    cout << "Length in cm: ";
    cin >> rec1.l;
    cout << "Width in cm: ";
    cin >> rec1.w;

    cout << "Area in cm^2: " << (rec1.l)*(rec1.w) << endl;
    cout << "Perimiter in cm: " << 2*(rec1.l + rec1.w) << endl;
    cout << "Continue? y/n: ";
    cin >> decision;

    while (decision != 'n') {
    cout << endl;
    cout << "Rectangle." << endl;
    cout << "Length in cm: ";
    cin >> rec1.l;
    cout << "Width in cm: ";
    cin >> rec1.w;

    cout << "Area in cm^2: " << (rec1.l)*(rec1.w) << endl;
    cout << "Perimiter in cm: " << 2*(rec1.l + rec1.w) << endl;

    cout << "Continue? y/n: ";
    cin >> decision;

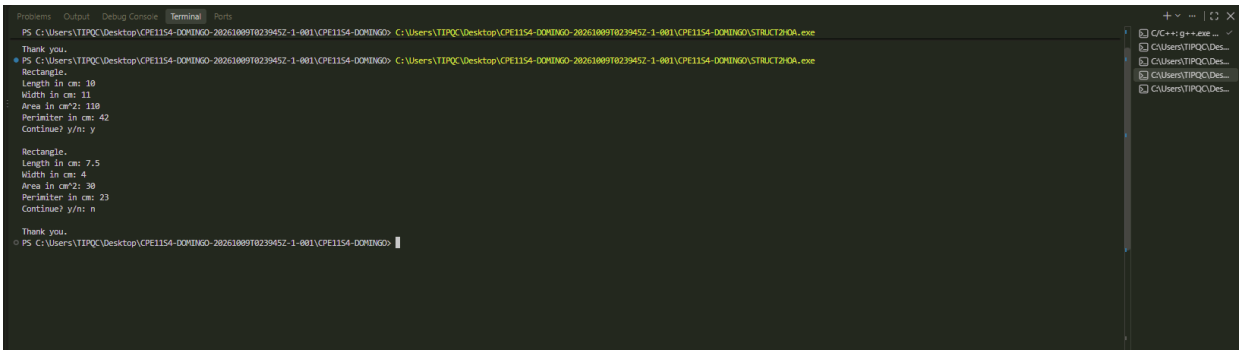
    }

    cout << endl;
    cout << "Thank you.";



    return 0;
}
```

OUTPUT



EXPLANATION: This program calculates the area and perimeter of a rectangle, instead of unnecessary hardcoding the program uses a data type called struct to represent the needed values for a rectangle's length and width. The variable rec1 is the one that inherits the elements within Rectangle which can be simultaneously for the length and width collection. It also has a bonus function of looping the process until the user enters the terminator variable.

Create a structure named Multiple to store an integer entered by the user and another integer x. Write a program that uses the structure to determine and display whether the first integer is a multiple of x.

INPUT

CPE11S4-DOMINGO > STRUCT3HOA.cpp > ...

```cpp
#include <iostream>
#include <string>
using namespace std;

struct Multiple {
    int num;
    int x;
};


int main () {
    char d;
    Multiple mult1;
    cout << "Multiple." << endl;
    cout << "Enter first integer: ";
    cin >> mult1.num;
    cout << "Enter another integer: ";
    cin >> mult1.x;

    if (mult1.x % mult1.num == 0) {
        cout << mult1.x << " is a multiple of " << mult1.num << endl;
    }
    else {
        cout << mult1.x << " is not a multiple of " << mult1.num << endl;
    }
    cout << "Continue? y/n: ";
    cin >> d;

    while (d != 'n') {
    cout << endl;
    cout << "Multiple." << endl;
    cout << "Enter first integer: ";
    cin >> mult1.num;
    cout << "Enter another integer: ";
    cin >> mult1.x;

    if (mult1.x % mult1.num == 0) {
        cout << mult1.x << " is a multiple of " << mult1.num << endl;
    }
    else {
        cout << mult1.x << " is not a multiple of " << mult1.num << endl;
    }
    cout << "Continue? y/n: ";
    cin >> d;
    }
    cout << endl;
    cout << "Thank you.";



    return 0;
}
//Create a structure named Multiple to store an integer entered by the user and another integer x. Write a program that uses the structure to determine and display whether the first integer is a multiple of x.
```

OUTPUT



EXPLANATION: This program gets the first integer from the user and another integer to check whether it is a multiple of the first integer, I used the modulo function to determine if the other number is a multiple of the first integer. Also added a bonus that loops the process until the user enters the terminator variable ‘n’.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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